Non-stationary models I

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ECMA 31150, Spring 2026

University of Chicago

Department of Economics

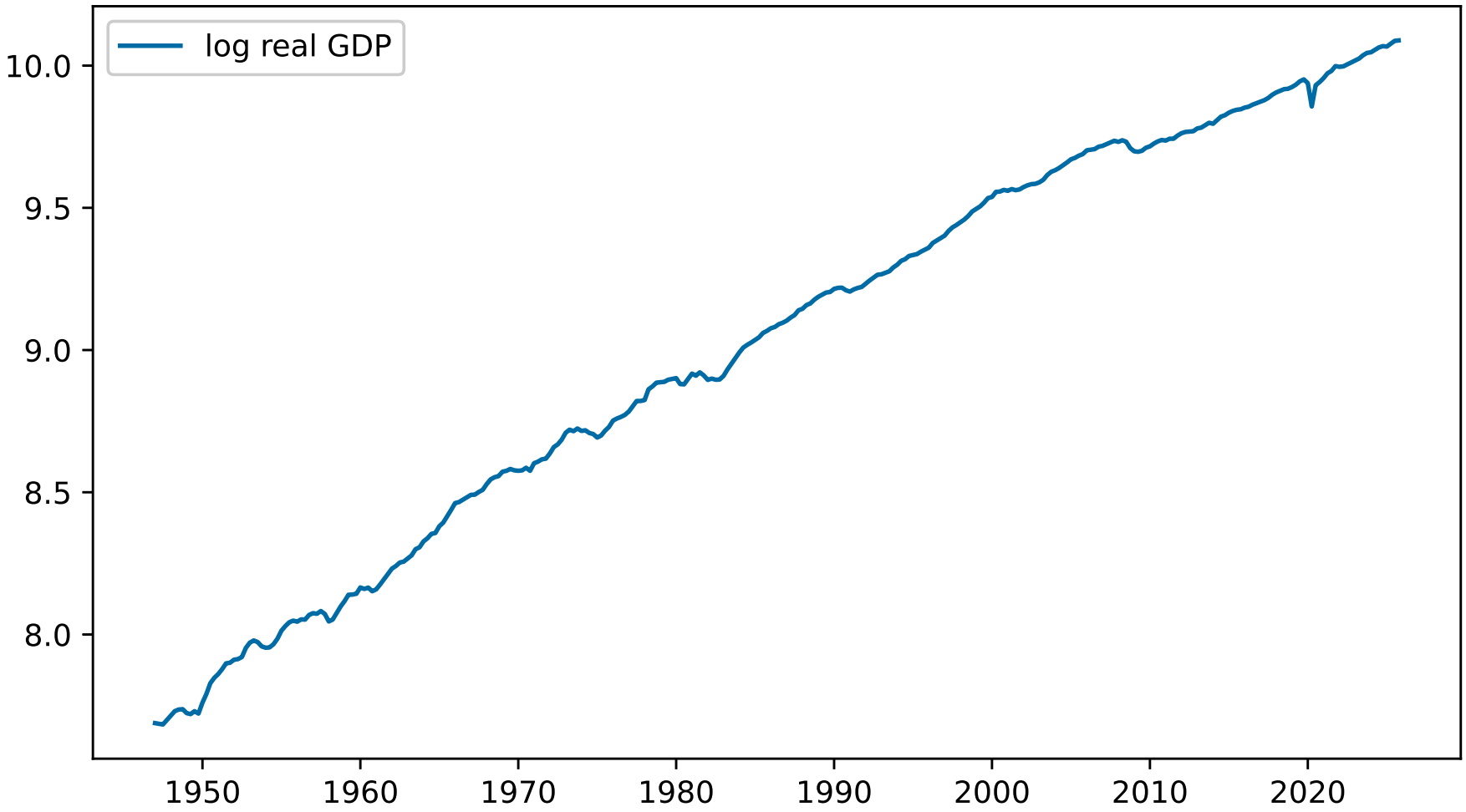
Mikkel Plagborg-Moller

Plan for this slide deck

1. Models for trending data
2. What happens if we try to estimate a non-stationary AR model by OLS?
3. How can we test the stationarity assumption?
4. Should we even try to categorize series as stationary or non-stationary?

Trending data

- As we've seen many times, lots of economic time series exhibit clear trends over time.
- This makes them **non-stationary**.
- So far we've primarily been dealing with this by transforming the data to stationarity, e.g., by differencing.
- But what if we care about the trends *per se*? And even if we don't, are we losing statistical power by transforming the data?


log real GDP
10.0
9.5
9.0
8.5
8.0
1950
1960
1970
1980
1990
2000
2010
2020

Trend stationarity

- One approach to trends is to model the data as arising from stationary fluctuations around a **deterministic time trend**:

$$y_t = \underbrace{\beta_0 + \beta_1 t}_{\text{deterministic}} + v_t,$$

  where $\{v_t\}$ is stationary and weakly dependent.

- We say that $\{y_t\}$ is **trend stationary**.

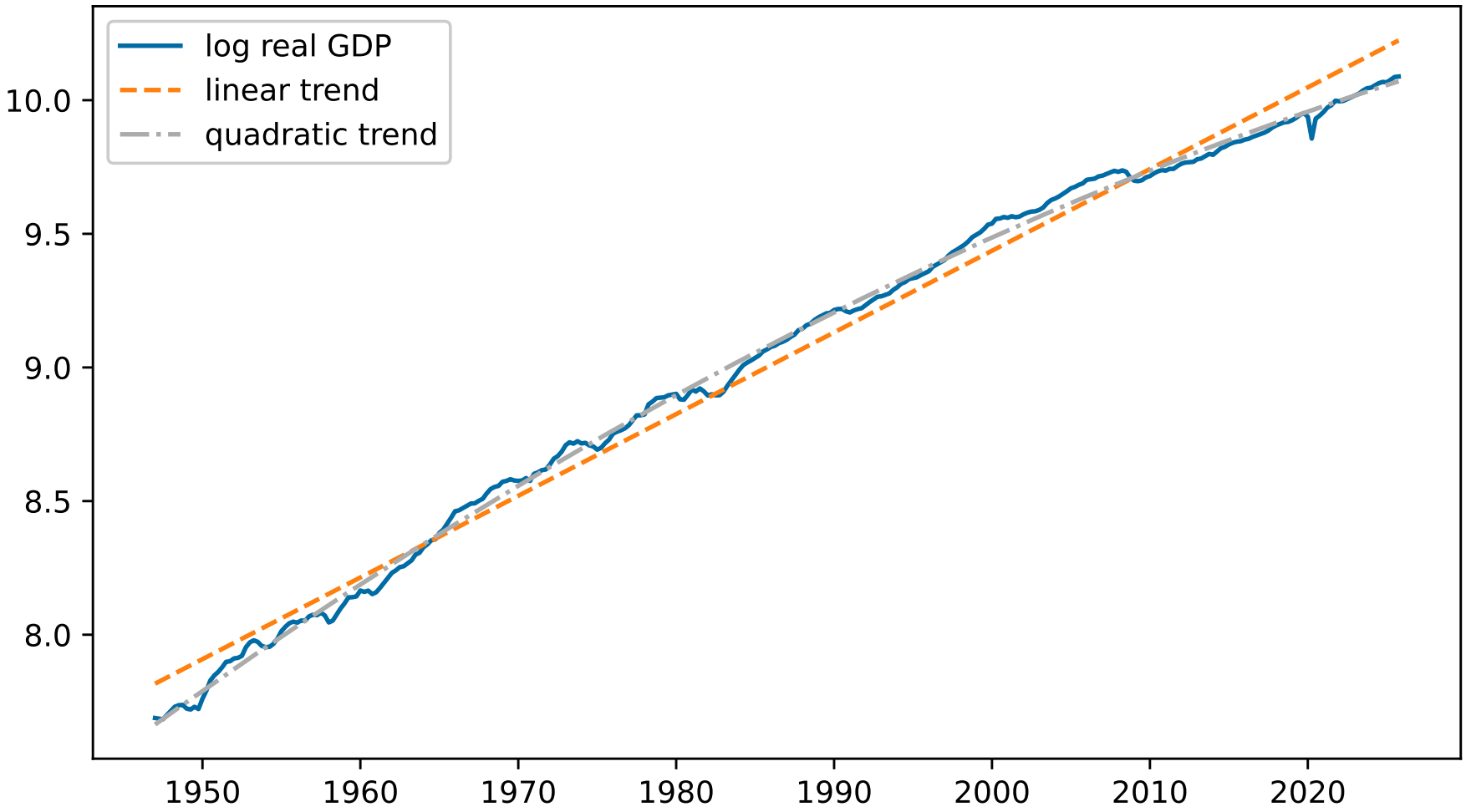
- The trend need not be linear:

$$y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots + \beta_p t^p + v_t.$$

```
# Trend regression
y = l_gdp.values
T = y.shape[0]
trend = np.arange(T)

# Linear
X_lin = sm.add_constant(trend)
res_lin = sm.OLS(y, X_lin).fit()

# Quadratic
X_quad = sm.add_constant(np.column_stack([trend, trend**2]))
res_quad = sm.OLS(y, X_quad).fit()
```


log real GDP
linear trend
quadratic trend
10.0
9.5
9.0
8.5
8.0
1950
1960
1970
1980
1990
2000
2010
2020

Trend stationarity: forecasting

- Optimal h-step forecast of a trend stationary series:

$$E[y_{T+h} \mid y_T, y_{T-1}, \dots] = \underbrace{\beta_0 + \beta_1(T+h) + \cdots + \beta_p(T+h)^p}_{\text{no uncertainty}} + E[v_{T+h} \mid y_T, y_{T-1}, \dots],$$

  with MSFE

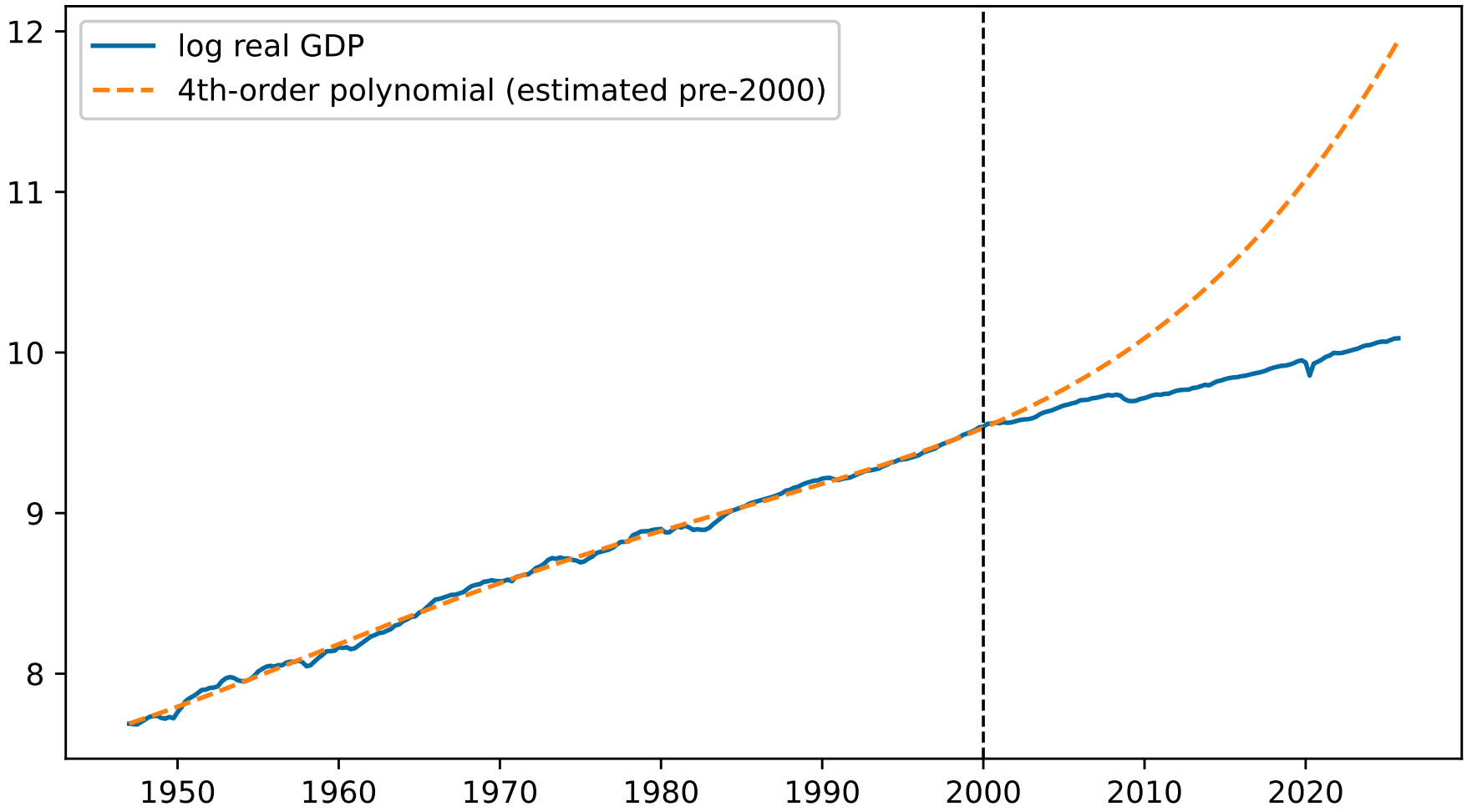
$$\text{Var}(y_{T+h} \mid y_T, y_{T-1}, \dots) = \text{Var}(v_{T+h} \mid y_T, y_{T-1}, \dots).$$

- The trend itself does not contribute at all to the MSFE (apart from parameter estimation error), since we can extrapolate out of sample based on the deterministic functional form.

Trend stationarity: forecasting (cont.)

$$\text{Var}(y_{T+h} \mid y_T, y_{T-1}, \dots) = \text{Var}(v_{T+h} \mid y_T, y_{T-1}, \dots)$$

- If $\{v_t\}$ is stationary and weakly dependent, then as the forecast horizon $h \to \infty$, we have $\text{Var}(v_{T+h} \mid y_T, y_{T-1}, \dots) \to \text{Var}(v_t)$.
- Hence, the MSFE for a trend stationary series $\{y_t\}$ stabilizes at long horizons $h \to \infty$, rather than growing without bound.
- This seems implausible – in practice, we're very uncertain about the level of real GDP 10 years from now, say.
- Trend stationary models are also prone to **overfitting**: with a high polynomial order p, we can fit trends well *in sample*, but may do very poorly *out of sample*.


log real GDP
4th-order polynomial (estimated pre-2000)
12
11
10
9
8
1950
1960
1970
1980
1990
2000
2010
2020

Stochastic trends

- Deterministic trends are unrealistic. We would like a model with a **stochastic trend**.
- A building block model is the **random walk**:

$$y_t = y_{t-1} + u_t, \quad u_t \overset{iid}{\sim} (0, \sigma^2).$$

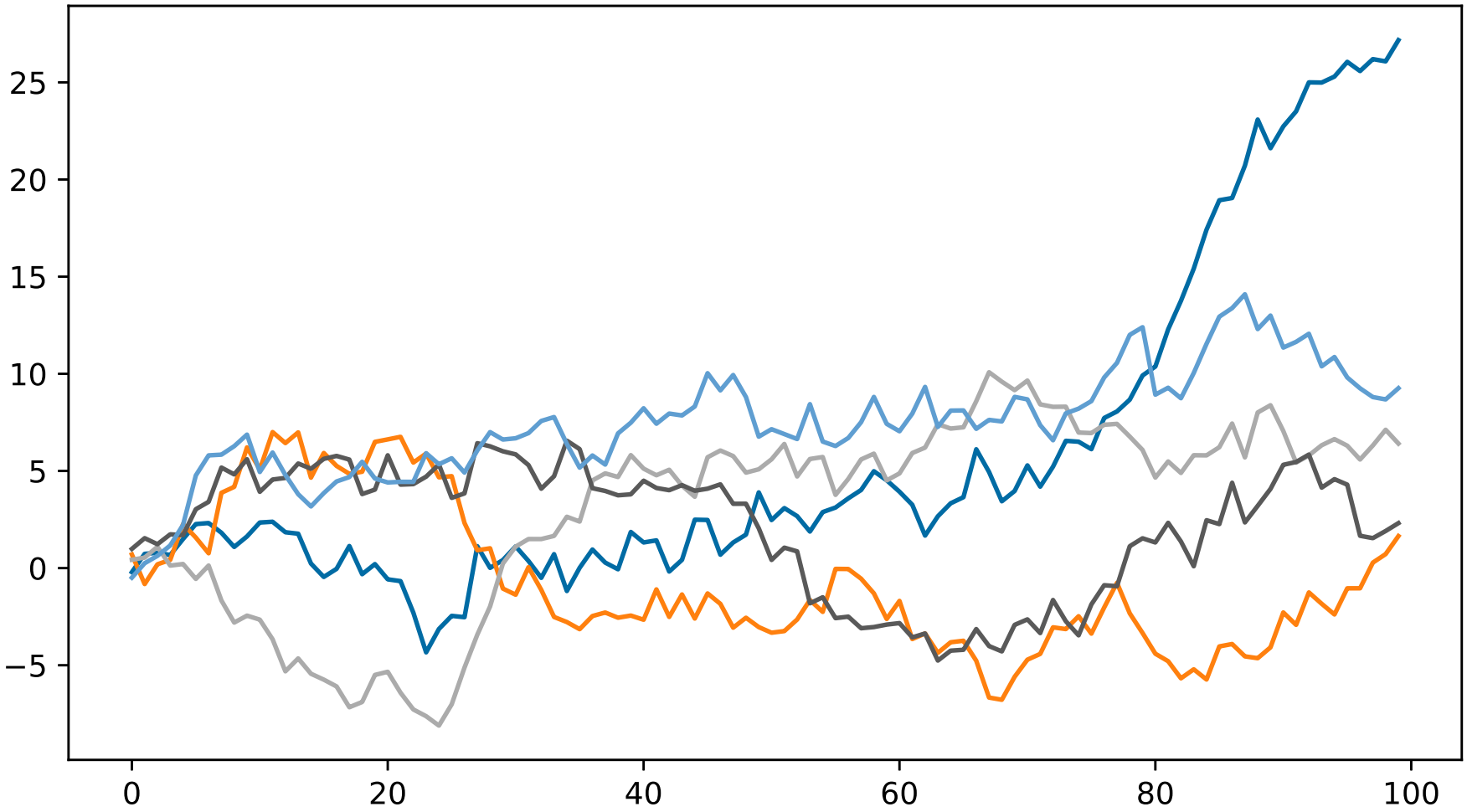
  This is an AR(1) model with $\rho = 1$.

- Note:

$$y_t = u_t + u_{t-1} + \cdots + u_1 + y_0 = y_0 + \sum_{s=1}^{t} u_s.$$

- If we treat the initial condition y_0 as a random variable independent of $\{u_t\}$, then

$$\text{Var}(y_t) = \text{Var}(y_0) + t\sigma^2 \to \infty \quad \text{as } t \to \infty.$$



5 sample paths of random walks with $u_t \overset{iid}{\sim} N(0,1)$ and $y_0 = 0$

Notice: some paths exhibit apparent trends, but not deterministic

Random walk: forecasting

$$y_t = y_{t-1} + u_t, \quad u_t \overset{iid}{\sim} (0, \sigma^2)$$

- For a random walk, the optimal forecast is "no change", both in the short and long run:

$$E[y_{T+h} \mid y_T, y_{T-1}, \dots] = E\left[y_T + \sum_{\ell=1}^{h} u_{T+\ell} \,\middle|\, y_T, y_{T-1}, \dots \right] = \boxed{y_T}.$$

- MSFE is linearly increasing in the forecast horizon h:

$$\text{Var}(y_{T+h} \mid y_T, y_{T-1}, \dots) = \text{Var}\left(\sum_{\ell=1}^{h} u_{T+\ell} \right) = \boxed{h\sigma^2}.$$

More realistic than trend stationary models.

Random walk with drift

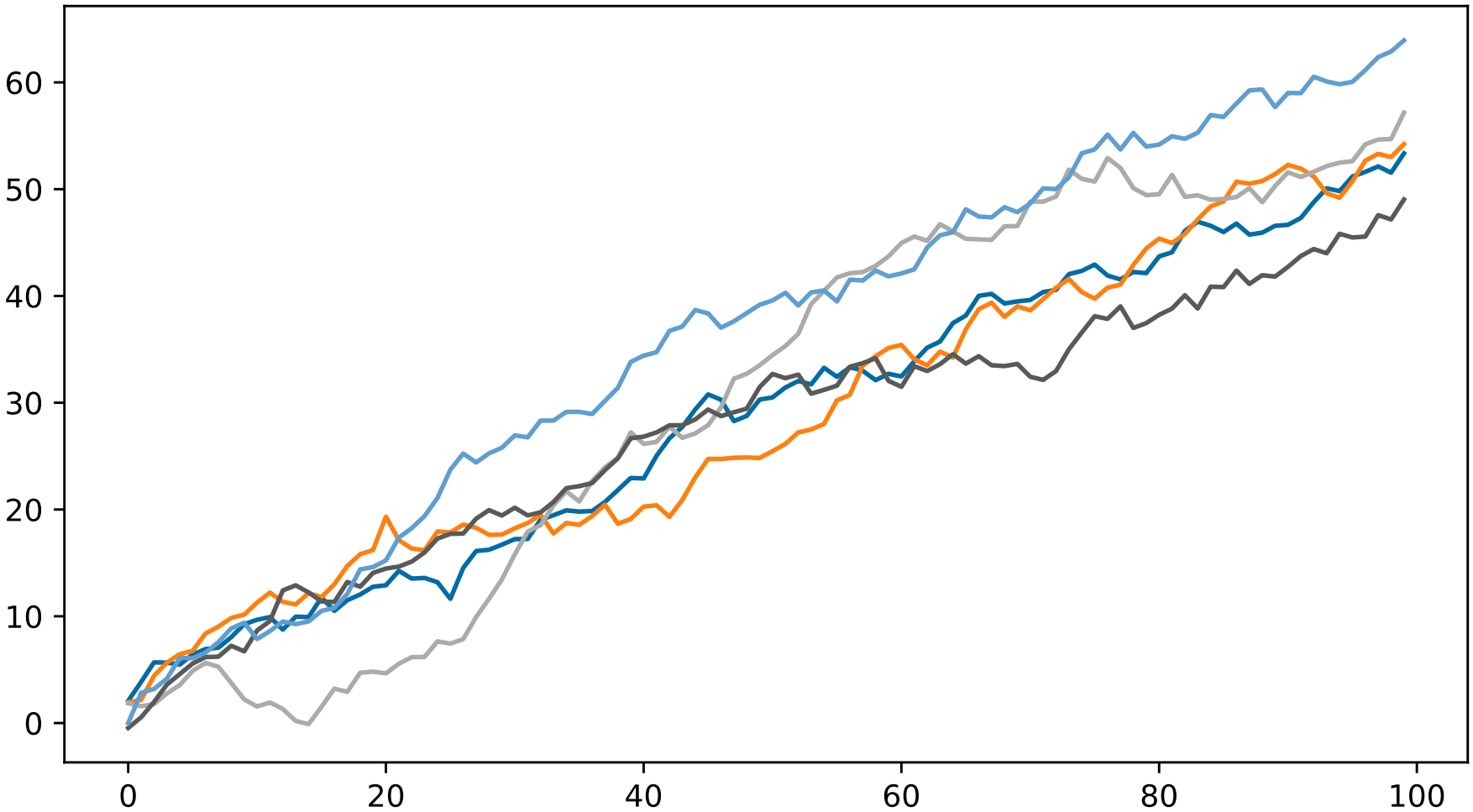
- The pure random walk is equally likely to exhibit increasing or decreasing stochastic trends.
- We can introduce a positive **drift** in $\{y_t\}$ by introducing an intercept:

$$y_t = c + y_{t-1} + u_t, \quad u_t \overset{iid}{\sim} (0, \sigma^2)$$

$$\implies \quad y_t = y_0 + \underbrace{ct}_{\text{deterministic trend}} + \underbrace{\sum_{s=1}^{t} u_s}_{\text{stochastic trend}} .$$

- Long-run forecasts remain very uncertain (exercise: verify):

$$\text{Var}(y_{T+h} \mid y_T, y_{T-1}, \dots) = \boxed{h\sigma^2}.$$



$$c = 0.5, \quad u_t \overset{iid}{\sim} N(0,1), \quad y_0 = 0$$

ARIMA

- The random walk with drift may be a good description of the long run. But we want a richer model of short-to-medium-run dynamics.
- Let's assume that $\{y_t\}$ is cov. stationary after transforming to first differences $\Delta y_t \equiv y_t - y_{t-1}$.
- Then we say that $\{y_t\}$ is **integrated of order 1** or **I(1)**.
  - If instead Δy_t is non-stationary but $\Delta^2 y_t \equiv \Delta(\Delta y_t)$ is cov. stationary, then we say that $\{y_t\}$ is I(2), and so on.
- Then we can apply our flexible ARMA modeling techniques to the first differences Δy_t:

$$\tilde{a}(L)(\Delta y_t - \mu) = \theta(L)u_t, \quad u_t \sim WN(\sigma^2).$$

- **ARIMA**: AutoRegressive Integrated Moving Average.

ARIMA (cont.)

$$\tilde{a}(L)(\Delta y_t - \mu) = \theta(L)u_t, \quad u_t \sim WN(\sigma^2),$$

$$\tilde{a}(L) \equiv 1 - \sum_{\ell=1}^{p-1} \tilde{a}_\ell L^\ell, \quad \theta(L) \equiv 1 + \sum_{\ell=1}^{q} \theta_\ell L^\ell.$$

- Assume $\tilde{a}(L)$ has all roots outside the unit circle, so $\{\Delta y_t\}$ is cov. stationary.
- Random walk with drift is a special case: $\tilde{a}(L) = 1$, $\theta(L) = 1$, $\mu = c$.
- The above equation does not fully pin down $y_1, y_2, \ldots$; we also need to specify an initial condition y_0. Then

$$y_t = y_0 + \sum_{s=1}^{t} \Delta y_s = y_0 + \underbrace{\mu t}_{\text{deterministic trend}} + \sum_{s=1}^{t} \underbrace{(\Delta y_s - \mu)}_{\text{stationary}}.$$

Autoregression with unit root

- Consider an ARIMA model with $\theta(L) = 1$. Then, since $\Delta \equiv 1 - L$,

$$\tilde{a}(L)(\Delta y_t - \mu) = u_t, \quad u_t \sim WN(\sigma^2)$$

$$\implies \quad a(L)y_t = c + u_t, \quad \text{where } a(L) \equiv (1 - L)\tilde{a}(L),\ c \equiv \tilde{a}(1)\mu.$$

- This is an AR(p) model (since we assumed $\tilde{a}(L)$ is of order $p - 1$).
- One of the roots of the AR polynomial $a(L)$ is equal to 1:

$$a(1) = (1 - 1)\tilde{a}(1) = 0.$$

  We call this an autoregression with a **unit root**.

- The random walk with drift is an AR(1) with a unit root:

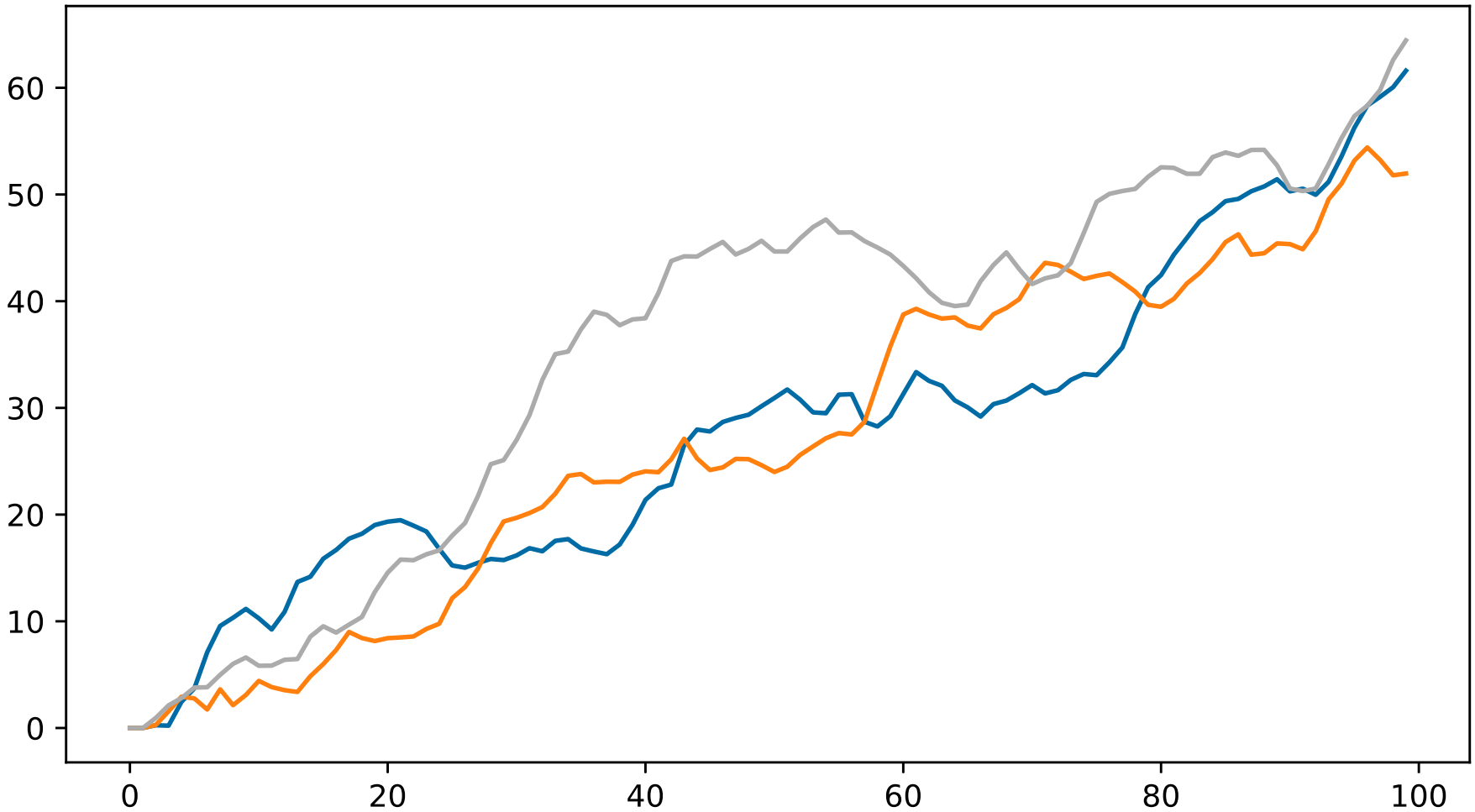
$$\underbrace{(1 - L)}_{a(L)} y_t = c + u_t \quad \implies \quad a(1) = 0.$$

Autoregression with unit root (cont.)

$$a(L)y_t = c + u_t, \quad a(L) \equiv 1 - \sum_{\ell=1}^{p} a_\ell L^\ell$$

- The AR(p) process has a unit root if and only if the sum of the autoregressive coefficients equals 1:

$$a(1) = 0 \iff 1 - \sum_{\ell=1}^{p} a_\ell \times 1^\ell = 0 \iff \boxed{\sum_{\ell=1}^{p} a_\ell = 1}.$$



$$y_t = 0.3 + 1.5y_{t-1} - 0.5y_{t-2} + u_t, \quad u_t \overset{iid}{\sim} N(0,1)$$

Estimation with stochastic trends

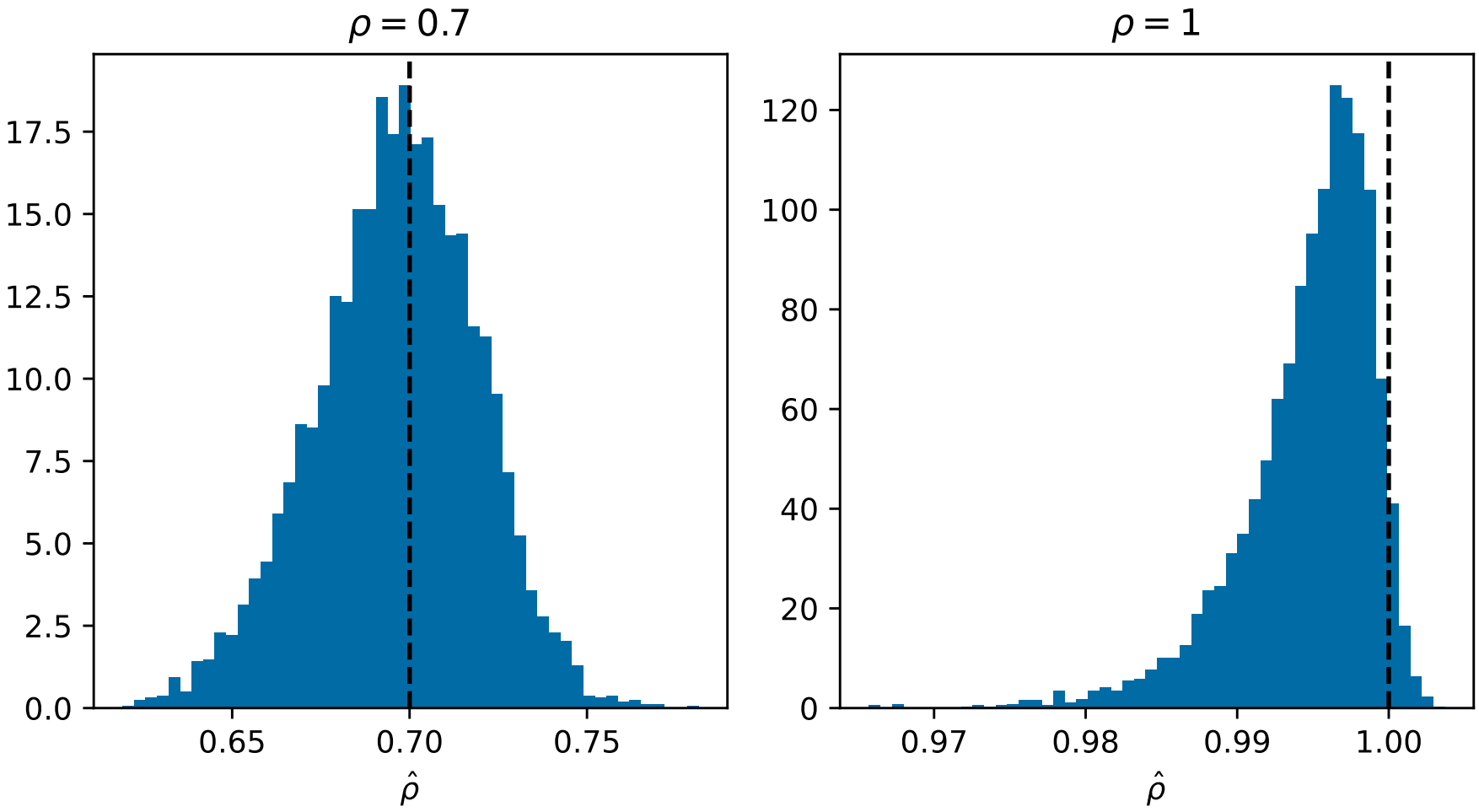
- If we are certain that a time series has a stochastic trend, then we can fit an AR/ARMA model in first differences.
- But sometimes we don't know *a priori* whether to model a series as trending or stationary (but persistent).
- We may want to *test* whether the series has a stochastic trend.

OLS estimation with a unit root

- Suppose we estimate an AR(1) model by OLS:

$$y_t = \hat{c} + \hat{\rho} y_{t-1} + \hat{u}_t.$$

- If the true AR(1) process has $|\rho| < 1$, then we know that $\hat{\rho}$ and associated t-statistic are normally distributed in large samples.
- But if the true process has a unit root, then this is no longer the case!
- Next slide: simulated sampling distributions of $\hat{\rho}$ when true ρ equals 0.7 and 1, respectively.



$$y_t = \rho y_{t-1} + u_t, \quad u_t \overset{iid}{\sim} N(0,1), \quad T = 1000, \quad \text{5000 simulations}$$

Notice the different axis scales

OLS estimation with a unit root (cont.)

- In the unit root case $\rho = 1$, the large-T sampling distribution of $\hat{\rho}$ is left-skewed and definitely not normal.
- The distribution is highly concentrated, though: $\text{Var}(\hat{\rho})$ is very small.
- Intuitively, we know that $\hat{\rho}$ is biased downward in finite samples. The bias is increasing in the persistence ρ. As $\rho \to 1$, the bias is no longer negligible relative to the (small) standard deviation of $\hat{\rho}$.
- $\hat{\rho}$ remains *consistent* for $\rho = 1$ as $T \to \infty$. In fact, it's **super-consistent**: its standard deviation decreases at rate $\boxed{1/T}$, not $1/\sqrt{T}$ as we're used to.
- Summary: when $\rho = 1$, $\hat{\rho}$ is highly precise, but its distribution is asymmetric. Invalidates conventional SE and confidence interval.

Unit root test

$$y_t = c + \rho y_{t-1} + u_t$$

- Suppose we want to test the null hypothesis that $\{y_t\}$ is a random walk without drift ($\rho = 1$, $c = 0$) against the stationary alternative ($|\rho| < 1$, c arbitrary).
- It's natural to compute the OLS estimator $\hat{\rho}$ and the t-statistic

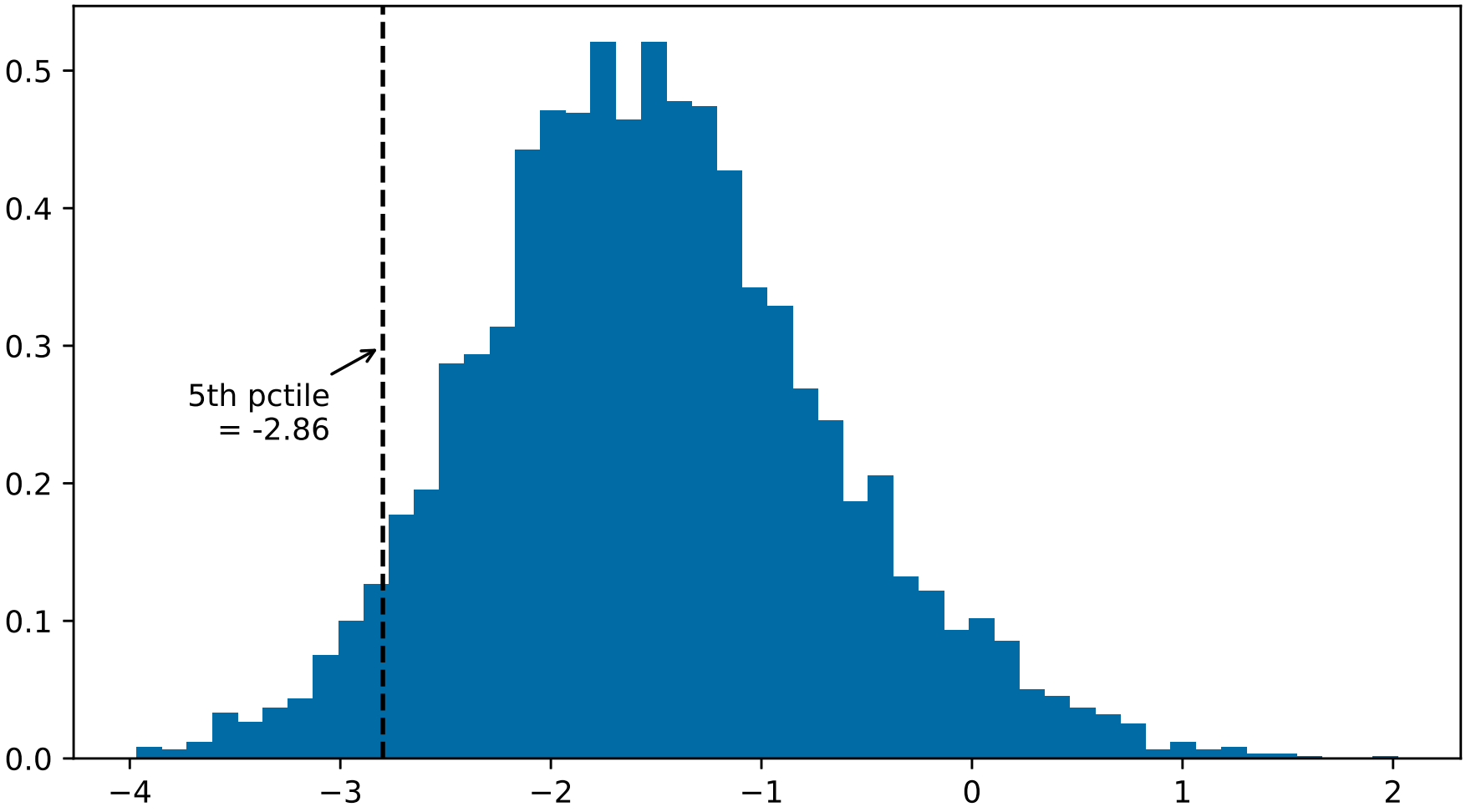
$$\hat{t} \equiv \frac{\hat{\rho} - 1}{SE(\hat{\rho})},$$

  where the denominator is the usual SE formula for OLS computed under *homoskedasticity* (will explain soon).
- If we find $\hat{t}$ to be very negative, we ought to reject the unit root hypothesis; otherwise we can't reject. How negative?

Unit root test (cont.)

$$\hat{t} \equiv \frac{\hat{\rho} - 1}{SE(\hat{\rho})}$$

- Under the null hypothesis $c = 0$ and $\rho = 1$, we've seen that the numerator has a non-normal distribution. Turns out that so does $\hat{t}$.
- But it can be shown that $\hat{t}$ converges under the null to another distribution, called the **Dickey-Fuller distribution**.
- This distribution does not depend on any unknown parameters. So we can use its 5th percentile as a critical value for a 5% test.
  - Surprisingly, this limiting distribution obtains also when $\{u_t\}$ is a heteroskedastic MDS, even though we used homoskedastic $SE(\hat{\rho})$.



$$y_t = y_{t-1} + u_t, \quad u_t \overset{iid}{\sim} N(0,1), \quad T = 1000, \quad \text{5000 simulations}$$

Unit root test: AR(p)

- Consider testing for a unit root in a more general AR(p) model

$$y_t = c + a_1 y_{t-1} + \cdots + a_p y_{t-p} + u_t.$$

- Exercise: show that we can rewrite this model as

$$y_t = c + \rho y_{t-1} + \gamma_1 \Delta y_{t-1} + \cdots + \gamma_{p-1} \Delta y_{t-p+1} + u_t,$$

  where $\rho \equiv \sum_{\ell=1}^{p} a_\ell$ and $\gamma_\ell \equiv -\sum_{j=\ell+1}^{p} a_j$.

- We've seen that we have a unit root if and only if $\rho \equiv \sum_{\ell=1}^{p} a_\ell = 1$.

Unit root test: AR(p) (cont.)

- **Augmented Dickey-Fuller** (ADF) t-test:
  1. Select lag length p (e.g., AIC).
  2. OLS regression:

$$y_t = \hat{c} + \hat{\rho} y_{t-1} + \hat{\gamma}_1 \Delta y_{t-1} + \cdots + \hat{\gamma}_{p-1} \Delta y_{t-p+1} + \hat{u}_t.$$

  3. Compute t-statistic with homoskedastic OLS SE:

$$\hat{t} \equiv \frac{\hat{\rho} - 1}{SE(\hat{\rho})}.$$

  4. Reject null hypothesis of unit root (without drift) if $\hat{t} < cv_\alpha$.
- It turns out that $\hat{t}$ has the same null distribution as in the AR(1) case earlier. So the 5% critical value $cv_{0.05}$ remains -2.86.
  - See Stock & Watson, chapter 15.7, for 1% and 10% crit. values.

Unit root test: AR(p) (cont.)

- Some papers/software use an alternative but equivalent formulation of the ADF t-test.
- Suppose we run an ADF regression, but with the dependent variable in first differences rather than levels:

$$\Delta y_t = \hat{c} + \hat{\delta} y_{t-1} + \hat{\gamma}_1 \Delta y_{t-1} + \cdots + \hat{\gamma}_{p-1} \Delta y_{t-p+1} + \hat{u}_t.$$

- By the linearity of OLS, $\hat{\delta} = \hat{\rho} - 1$ (exactly).
- So with this regression specification, the ADF test statistic should be for the hypothesis $\delta = 0$ (not 1).
- Critical value unchanged, of course.

```
# Manual ADF test
p = 4
dy = np.diff(y) # GDP growth

X_adf = pd.DataFrame(
    np.column_stack([y[p-1:-1], lagmat(dy, p-1, trim='both')]),
    columns=['y_lag1'] + [f'Dy_lag{k}' for k in range(1, p)],
)
X_adf = sm.add_constant(X_adf)
res_adf = sm.OLS(pd.Series(dy[p-1:], name='dy'), X_adf).fit()
print(res_adf.summary())
```

OLS Regression Results

Dep. Variable:	dy	R-squared:	0.043
Model:	OLS	Adj. R-squared:	0.031
Method:	Least Squares	F-statistic:	3.490
Date:	Fri, 15 May 2026	Prob (F-statistic):	0.00833
Time:	15:39:35	Log-Likelihood:	969.17
No. Observations:	312	AIC:	-1928.
Df Residuals:	307	BIC:	-1910.
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0248	0.008	2.949	0.003	0.008	0.041
y_lag1	-0.0020	0.001	-2.230	0.026	-0.004	-0.000
Dy_lag1	0.1092	0.057	1.919	0.056	-0.003	0.221
Dy_lag2	0.0886	0.057	1.556	0.121	-0.023	0.201
Dy_lag3	-0.0418	0.057	-0.738	0.461	-0.153	0.070

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Df Model:			
Covariance Type:			

$\hat{t} > -2.86$, so can't reject unit root

	coef	std err	t	P>|t|	[0.025	0.975]
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y_lag1	-0.0020	0.001	-2.230	0.026	-0.004	-0.000
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Ignore this!
(assumes normality)

```
from statsmodels.tsa.stattools import adfuller

print('Null: RW no drift, alternative: stationary')
adf_stat, p_val, *_ = adfuller(y, maxlag=p-1, regression='c', autolag=None)
print(f't-stat: {adf_stat:.4f}, p-value: {p_val:.4f}')
```

Results:

```
Null: RW no drift, alternative: stationary
t-stat: -2.2305, p-value: 0.1954
```

```
from statsmodels.tsa.stattools import adfuller

print('Null: RW no drift, alternative: stationary')
adf_stat, p_val, *_ = adfuller(y, maxlag=p-1, regression='c', autolag=None)
print(f't-stat: {adf_stat:.4f}, p-value: {p_val:.4f}')
```

Results:

Same as previous slide

```
Null: RW no drift, alternative: stationary
t-stat: -2.2305, p-value: 0.1954
```

Note: `maxlag` = number of lagged first differences $\Delta y_{t-\ell}$ in the regression

ADF test with clearly trending data

- The ADF test on the preceding slides tests the null hypothesis of a random walk *without* drift against the alternative of *stationarity*.
- This test does not make much sense if the data has a very clear positive or negative time trend.
- Instead, consider testing the null hypothesis of a random walk *with* drift against the alternative of *trend* stationarity. Then we need to add a time trend to the ADF regression:

$$\Delta y_t = \hat{c} + \hat{\zeta} t + \hat{\delta} y_{t-1} + \hat{\gamma}_1 \Delta y_{t-1} + \cdots + \hat{\gamma}_{p-1} \Delta y_{t-p+1} + \hat{u}_t.$$

- Compute the t-statistic for $\delta = 0$. Limiting distribution under the null differs from before. Critical values: Stock & Watson, chapter 15.7.

```
print('Null: RW w/drift, alternative: trend stationary')
adf_stat2, p_val2, *_ = adfuller(y, maxlag=p-1, regression='ct', autolag=None)
print(f't-stat: {adf_stat2:.4f}, p-value: {p_val2:.4f}')
```

Results:

```
Null: RW w/drift, alternative: trend stationary
t-stat: -1.6259, p-value: 0.7821
```

`regression='c'`: no time trend, just intercept
`regression='ct'`: include time trend

Near-unit roots

- In data of a given sample size T, there is little difference between an AR(p) process with $\rho = 1$ and one with $\rho = 0.999$, say.
  - Recall $\rho \equiv \sum_{\ell=1}^{p} a_\ell$.
- So if we have a root very *close* to 1, normality-based statistical procedures could behave poorly (unless T is astronomically large).
- The case $\rho \approx 1$ requires as much attention as $\rho = 1$.
- You might hope that the bootstrap could help us adapt to the magnitude of ρ, but unfortunately not when $\rho \approx 1$.
  - Intuition: the bootstrap generates fake data based on the severely downward-biased OLS estimate $\hat{\rho}$ of ρ. So the bootstrap is "too optimistic" about how normally distributed $\hat{\rho}$ is.

Dealing with unit roots in practice

- When researchers are unsure about how to model a potentially trending series $\{y_t\}$, it's common to proceed in steps:
  - Run ADF test.
  - Reject unit root $\Rightarrow$ estimate stationary model for $\{y_t\}$.
  - Fail to reject unit root $\Rightarrow$ estimate stationary model for $\{\Delta y_t\}$.
- But the first-step **pre-test** will sometimes commit a type-I or type-II error, which can mess up later inference.

Dealing with unit roots in practice (cont.)

- If we don't care about the presence of a unit root *per se*, it's often better to skip the pre-test and just estimate the AR model in *levels* (or log levels):

$$y_t = \hat{c} + \hat{a}_1 y_{t-1} + \cdots + \hat{a}_p y_{t-p} + \hat{u}_t.$$

- If the true process has a unit root, then $\hat{\rho} \equiv \hat{a}_1 + \cdots + \hat{a}_p$ has a non-normal distribution, as we've seen.

- But if the lag order $p > 1$, the individual coefficients $\hat{a}_\ell$ typically have (asymptotic) normal distributions, so we can use conventional normality-based tests and confidence intervals for these.

- Moreover, impulse responses – which are functions of the coefficients – are also usually normally distributed.

- These conclusions extend to VARs: just use variables in (log) levels.

Dealing with unit roots in practice (cont.)

- A Bayesian who is confident in her prior need not be concerned about unit roots.
- There is nothing in our likelihood calculation that depends on stationarity. So posterior inference can proceed as usual.
- But results could be quite sensitive to the choice of prior if some roots are close to 1.

What did we learn?

- Trend stationarity
- Stochastic trend, random walk
- AR model with unit root
  - Super-consistent OLS estimator
  - Non-normal estimator and t-statistic
- ADF test for unit root
- Often best to avoid pre-tests and just estimate the AR/VAR in (log) levels

Looking ahead

- Modeling multiple time series with common trends
- Testing for structural breaks
- Removing trends and seasonality from data